

Title

Titanic Dataset - Exploratory Data Analysis Report

Objective

Analyze the Titanic dataset using statistical summaries and visualizations to identify patterns and relationships.

Tools

Python, Pandas, Matplotlib, Seaborn, Jupyter Notebook

Analysis

Performed data inspection with `info()`, `describe()`, checked missing values, created histograms, boxplots, scatter plots, pairplot, and a correlation heatmap.

Key Findings

Female passengers had higher survival rates. First-class passengers survived more often. Fare showed a positive relationship with survival. The dataset contained missing values, mainly in the Age column, which were handled using the median.

Conclusion

EDA revealed important trends and relationships that can support future predictive modeling.